Bavireddy Jaswanth Rishikesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.25
Sri Chaitanya Educational Institutions	Class XII (TSBIE)	2024	96.90%
Geetanjali High School	Class X (CBSE)	2022	98.40%

EXPERIENCE

Texas Instruments | BYTE Program (Certified)

- Worked on analog fundamentals; qualified the certification test and was shortlisted among the top 3 participants.

Core Team Member - IITH Contingent | Inter-IIT Tech Meet 14.0

- Supported senior team members in the behavioral modeling and evaluation of Delta-Sigma modulator using Simulink
- Conducted simulation studies, analyzed performance trade-offs, and contributed to technical reports and presentations, helping the team achieve 6th place nationally.

Teaching Assistant | Circuits and Network Analysis EE1101

- Developed & delivered tutorials on Ngspice; designed assignments and conducted doubt solving sessions on circuits

Core Member | Elektronica - Electronics and Signal Processing Club, IITH

- Delivered hands-on sessions on FPGA; Worked on RISC V Processor RTL design and implementation on PYNQ-Z2

PROJECTS

Analog Circuit Design and Simulation (LTSpice)

- Practical Op-amp integrator (10 kHz-800 kHz UGB) and differentiator (DC-10 kHz)
- Schmitt trigger with V_{th} = 200 mV; Square wave, Triangular and Sawtooth waveform generator with f_o = 10 kHz.
- RC phase shift (3-stage) and Wien bridge oscillators with f_o = 50 kHz.
- Second-order Sallen-Key active filters with cutoff frequencies of 72.3 Hz (HPF) and 15.9 kHz (LPF)
- Common source amplifier of small signal gain 25 with DC bias current of 0.1 mA.

Semiconductor Devices Lab

- Characterized the 1N4007 diode by extracting Is, Vt, and ideality factor from experimental I-V measurements.
- Characterized solar cell I-V and C-V to determine conversion efficiency and junction doping concentrations
- Characterized the transfer and output plots of 2N7000 MOSFET, extracting Vth, gm, subthreshold slope, and Ron

Drift-Diffusion based MOSCAP Simulator

- Developed a transient 1D MOS capacitor simulator to extract C-V characteristics under varying illumination and carrier generation conditions.
- Implemented Scharfetter-Gummel discretization, Crank-Nicolson scheme for the carrier continuity equation, 1D Poisson solver, and self-consistent Gummel iterations.

8-bit Breadboard Computer

- Built an 8-bit breadboard computer implementing an ALU, registers, RAM, clock, and microcode-based control logic
- Implemented the complete fetch-decode-execute cycle with EEPROM-based microcode, enabling program execution through manual assembly programming.

RTL Design and Verification

- Designed a 5-stage pipelined RISC-V (RV32I) processor in Verilog; verified functionality using Verilog testbenches.
- Implemented the processor on the PYNQ-Z2 (Zynq-7000 SoC), integrating an AXI4-Lite memory-mapped interface for PS-PL communication, run-time program loading, and hardware validation.

Custom gaming controller

- Developed firmware for ESP32 BLE controller integrating joysticks, D-pad, triggers, motion sensing, and LEDs
- Implemented joystick calibration, deadzone compensation, axis mapping, input smoothing, and gyroscope fusion to improve control accuracy and responsiveness.

SCHOLASTIC ACHIEVEMENTS

- Secured A+ in 14 credits (8 courses) in first 3 semesters; A+ grade is typically given to only top 2% of the class

SKILLS

Programming	C | Python | Verilog | Bash	MCUs & FPGA	Arduino | ESP32 | Pynq-Z2
Tools	KiCad | LTspice | ngspice | MATLAB | Vivado | Xilinx SDK | Linux | Docker | Git		

RELEVANT COURSES

VLSI Courses : Circuits and Network Analysis, Control Theory for Circuits, Analog Electronics, Signals and Systems, Electrical Circuits Lab, Semiconductor Device Fundamentals, Introduction to CMOS Processing, Analog Lab, Semiconductor Devices Lab, Electronic System Design Project Lab, Digital Systems, Digital Systems Lab, Intro to IC Design and Technology, Productivity Tools for IC Design, Microprocessor & Computer Architecture, Electromagnetism

EXTRACURRICULAR

- Participated in CeNSE Winter School on Semiconductor Technology conducted by CeNSE - IISc, Bengaluru

- Completed 90+ hours of NSS service, delivering educational support to government school children, participating in anti-drug awareness campaigns, and contributing to blood donation and community welfare initiatives outside campus